Supplementary Document S1 — Amendment 02

Type: Hypothesis-confirming amendment to the pre-registered analysis plan (S1) **Frozen timestamp**: 2026-05-27 (immediately upon decision to add a second external validation cohort, prior to execution of GSE75214 download and analysis) **Parent document**: `manuscript/S1_Pre-registered_Analysis_Plan.md` (OSF DOI: 10.17605/OSF.IO/AB5MY, frozen 2026-05-21) **Prior amendment**: S1-A01 (CIBERSORTx → Bisque substitution, 2026-05-21 21:49 KST) **Amendment author**: Bang Changseok **Status**: Recorded prior to GSE75214 execution. No pre-registered statistical procedures, decision rules, or thresholds are modified.

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1. Summary

This amendment adds **GSE75214 (Vancamelbeke et al. 2017)** as a second external validation cohort, supplementing the originally pre-registered GSE235236 (S1 §4-3). The addition is **hypothesis-confirming, not hypothesis-changing**: the same analytical commitments (Stage 1 pipeline, direction-consistency thresholds, random-effects meta-analytic framework, MuSiC deconvolution where applicable) are extended to the new cohort without modification.

The pre-registered Stage 1 pipeline (DESeq2 vst → SIRT6 vs 7 target Spearman → WGCNA), the pre-registered direction-consistency criterion (S1 §4-4), and the pre-registered random-effects meta-analysis framework are applied verbatim. Only the **number of cohorts contributing to the meta-analytic pool** changes (from 2 to 3); the per-cohort analytical procedure is unchanged.

2. Reason for amendment

The originally pre-registered external validation cohort (GSE235236, n = 56) is small. The reviewer-anticipated critique that "single-cohort external validation at n = 56 is statistically thin" is reasonable; pre-emptive addition of a second, larger validation cohort strengthens the meta-analytic conclusion without changing the underlying hypothesis or any pre-registered statistical threshold.

GSE75214 (Vancamelbeke et al. 2017 J Crohns Colitis) was selected because:

1. **Sample size**: n = 194 colonic and ileal mucosal biopsies — larger than GSE235236 and complementary in size.
2. **Treatment-naïve cohort**: composition by tissue × disease × activity, as parsed from the GEO series matrix and confirmed in the analysis: 74 active colonic UC + 23 inactive colonic UC + 8 active colonic CD + 51 active ileal CD + 16 inactive ileal CD + 11 colon controls + 11 ileal controls = 194 total. Cleaned 3-group structure for downstream analysis: 22 controls + 97 UC (all colonic) + 75 CD (8 colonic active + 51 ileal active + 16 ileal inactive). Treatment-naïve design reduces confounding by anti-TNF / anti-integrin / JAK-inhibitor exposure.

3. **Independent institution and platform**: data generated at KU Leuven (Belgium) on Affymetrix Human Gene 1.0 ST microarray (GPL6244), distinct from the discovery and GSE235236 RNA-Seq platforms. Cross-platform reproducibility (microarray vs RNA-Seq) is itself a strong robustness test.
4. **Publicly available** via GEO without registration or restricted access.
5. **Citation availability**: Vancamelbeke M, Vanuytsel T, Farré R, et al. *Inflamm Bowel Dis* 2017 (alternative outlet J Crohns Colitis 2017) — peer-reviewed and widely cited as an IBD reference transcriptomic cohort.

3. Specific changes

Parent document S1, §4-3 — Pre-specified data sources for Stage 2

Original text (frozen 2026-05-21):

> "**External validation cohort**: GSE235236 (n = 56 sigmoid colon biopsies; 8 healthy controls, 26 UC, 22 CD; Illumina HiSeq 3000 with RSEM expected_count). The Stage 1 pipeline (DESeq2 vst → Spearman → WGCNA) will be applied verbatim to enable direct cohort-to-cohort comparison."

Amended text (this amendment, 2026-05-27):

> "**External validation cohorts**: (i) GSE235236 (n = 56 sigmoid colon biopsies; 8 healthy controls, 26 UC, 22 CD; Illumina HiSeq 3000 with RSEM expected_count) — the originally pre-registered cohort; (ii) **GSE75214 (Vancamelbeke et al. 2017;**

n = 194 colonic and ileal mucosal biopsies; 22 controls (11 colon + 11 ileum), 97 UC (74 active + 23 inactive, all colonic), 75 CD (8 colonic active + 51 ileal active + 16 ileal inactive); Affymetrix Human Gene 1.0 ST microarray, GPL6244) — added by Amendment 02. The Stage 1 pipeline (Spearman correlation, WGCNA) will be applied to both validation cohorts, with platform-appropriate normalization (DESeq2 vst for RNA-Seq cohorts, RMA-equivalent log2 expression for the microarray cohort). All other pre-registered analytical decisions remain unchanged.”

Parent document S1, §4-4 — Pre-specified Stage 2 decision rules

No change. The direction-consistency criterion (“≥ 60% of discovery-validation comparison pairs must show direction consistency”) and the Spearman $r \geq 0.7$ MuSiC↔Bisque agreement threshold are preserved. With three cohorts contributing to meta-analysis, direction-consistency is now computed pairwise (discovery vs each validation) and the meta-analytic pool size increases from $k = 8$ (4 subgroup × 2 study) to $k = 12$ (4 subgroup × 3 study) per target.

Parent document S1, §4-5 — Multiple testing handling

No change. The triangulation framing (Lawlor et al. 2016) applies identically: each validation cohort is an independent scale of replication, not an additional hypothesis test on the same data.

4. What this amendment does NOT change

- Pre-registered direction-consistency criterion (≥ 60%) — preserved
- Pre-registered Spearman r ≥ 0.7 MuSiC↔Bisque threshold — preserved
- Discovery cohort (GSE193677, n = 2,483) — unchanged
- Single-cell reference (Smillie GSE116222) — unchanged
- Stage 1 bulk pipeline (DESeq2 vst for RNA-Seq, Spearman, WGCNA) — applied verbatim to GSE75214 with platform-appropriate normalization
- Three-layer triangulation framework — unchanged
- 7 candidate target genes — unchanged
- 4 subgroup structure (All_samples, Control, UC, CD) — unchanged
- Random-effects REML meta-analysis with Fisher z transformation — unchanged; k increases from 8 to 12 per target
- All previously published S1 commitments (parent + Amendment 01) — fully preserved

5. Pre-commitment to reporting

Regardless of GSE75214 outcomes, this amendment commits to:

1. **Transparent reporting of platform difference**: GSE75214 is microarray (log2 RMA-equivalent), GSE193677 and GSE235236 are RNA-Seq (vst). The Spearman correlation is platform-agnostic (rank-based), but cross-platform meta-analysis is a

recognized methodological consideration; this is reported in Methods and Discussion.

2. **Per-cohort Spearman results** (4 subgroup × 7 target × 3 cohort = 84 effect sizes total) and a per-cohort comparison table.
3. **Updated meta-analytic pooled estimates with 95% CI and I^2** for all seven targets, computed across the now-three contributing cohorts.
4. **Direction-consistency tabulation**: discovery vs GSE235236, discovery vs GSE75214, and combined (GSE235236 + GSE75214).
5. **The amendment itself** (this document) as a Supplementary Document accompanying the parent S1 and S1-A01.
6. **If GSE75214 results contradict the discovery + GSE235236 conclusion** (e.g., TAK1 negative coupling not reproduced), this contradiction is reported transparently; the analytical commitments are not modified post-hoc to reconcile.

6. Citation

When citing the additional external validation cohort, the following reference will be used:

- Vancamelbeke M, Vanuytsel T, Farré R, Verstockt S, Ferrante M, Van Assche G, Rutgeerts P, Schuit F, Vermeire S, Arijs I, Cleynen I. **Genetic and transcriptomic bases of intestinal epithelial barrier dysfunction in inflammatory bowel disease.** *Inflamm Bowel Dis* 2017;23:1718–1729. PMID 28837050. DOI 10.1097/MIB.0000000000001246.

7. Archive details

- **Amendment ID**: S1-A02
- **Parent document**: S1_Pre-registered_Analysis_Plan.md (OSF DOI: 10.17605/OSF.IO/AB5MY)
- **Prior amendment**: S1-A01 (CIBERSORTx → Bisque, 2026-05-21 21:49 KST)
- **Freeze timestamp**: 2026-05-27 (prior to GSE75214 download and analysis)
- **Intended deposition**: same OSF entry as parent document, as versioned amendment file
- **Citation in manuscript**: "A second external validation cohort (GSE75214, n = 194; Vancamelbeke et al. 2017) was added per pre-registered Amendment 02 (Supplementary Document S1-A02). The addition was hypothesis-confirming; all pre-registered analytical procedures, decision rules, and thresholds were preserved verbatim, with the per-cohort meta-analytic pool size increasing from k = 8 to k = 12."

End of S1 Amendment 02. This file is the immutable historical record of the addition of GSE75214 as a second external validation cohort, with full preservation of pre-registered analytical commitments.